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$$\int \sin x \ln(\sin x)\, dx = -\int \ln(\sin x)\, d(\cos x) = -\ln(\sin x)\cos x + \int \frac{\cos^2 x}{\sin x} dx$$

$$= -\ln(\sin x)\cos x + \int \frac{\cos^2 x \sin x}{\sin^2 x} dx = -\ln(\sin x)\cos x - \int \frac{\cos^2 x}{1-\cos^2 x} d(\cos x)$$

$$\overset{\cos x = u}{=} -\ln(\sin x)\, u + \int \frac{u^2}{u^2-1} du = -\ln(\sin x)\, u + \int \left(\frac{1}{u^2-1} + 1\right) du$$

$$\Rightarrow \frac{1}{(u+1)(u-1)} = \frac{A}{u-1} + \frac{B}{u+1}$$

$$\Rightarrow A = \frac{1}{2}, B = -\frac{1}{2}$$

$$= -\ln(\sin x)\, u + \frac{\ln|u-1|}{2} - \frac{\ln|u+1|}{2} + u + c$$

$$\overset{u = \cos x}{=} -\ln(\sin x)\cos x + \frac{\ln|\cos x - 1|}{2} - \frac{\ln|\cos x + 1|}{2} + \cos x + c$$

References